Literature Review

Previous Research • Methods • Project Decisions • Limitations

DOCUMENT DETAILS	
Project	Telecom Customer Retention Intelligence & Churn Risk Prediction
Project Name	Churn Radar
Track	Data Analysis
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Team Leader	Judy
Deadline	16 October 2026
Document	Document 02 — Literature Review

Dataset: IBM Telco Customer Churn — 21-column CSV version (7,043 records).

1. Project and Dataset

Our project uses the 21-column Telco-Customer-Churn.csv version, which contains 7,043 customer records. The wider IBM Cognos sample describes a fictional telecommunications provider and defines Churn as whether a customer left during the previous month. We use the CSV version consistently throughout this project.

We investigate which recorded customer and service characteristics are associated with churn and evaluate how well machine-learning models identify customers labelled as churned.

2. Previous Research on Telecom Churn

Huang, Kechadi and Buckley (2012) studied churn prediction using landline telecommunications data and compared seven prediction approaches. Their work examines the use of customer and service characteristics in churn prediction.

Verbeke et al. (2012) presented a profit-driven approach to telecom churn evaluation. Their method ranks customers by predicted churn probability and estimates the potential value of targeting a portion of them for retention. Our project reports model metrics and error counts; financial return is outside our current analysis because campaign costs and offer outcomes are not included in the dataset.

Vafeiadis et al. (2015) compared machine-learning techniques for customer churn prediction, including boosting methods. Ahmad, Jafar and Aljoumaa (2019) studied telecom churn prediction using machine learning on a big-data platform. These studies provide examples of model comparison and data-processing approaches, while their datasets differ from the one used in our project.

3. Methods Used

Method	Purpose in Churn Radar
Exploratory and statistical analysis	We describe customer groups, compare churn rates and test selected relationships.
Logistic Regression	We use Logistic Regression as an interpretable baseline model.
Random Forest	We use Random Forest to model nonlinear relationships and interactions between customer characteristics.
Evaluation metrics	We report confusion matrix, precision, recall, F1 and ROC-AUC. ROC curves summarize the trade-off between true-positive and false-positive rates. Precision–recall analysis adds information about positive-class performance.
Class imbalance	We compare class weighting with the baseline and use SMOTE only if needed, applying it only to training data.

4. Research Findings Applied to Our Project

Research Finding	Our Project Approach
Model choice and features influence prediction results.	We compare Logistic Regression and Random Forest using the same split.
Prediction metrics alone do not establish campaign value.	We report error counts and avoid financial-return claims without campaign data.
ROC and precision–recall curves describe different aspects of model performance.	We report both where available, alongside class-specific metrics.
Class imbalance can affect churn detection.	We check class proportions and compare class weighting before considering SMOTE.

5. Improvements and Limitations

We select the classification threshold using validation data and report recall, precision and the false-positive trade-off. We use feature importance or SHAP to interpret the selected model, describing findings as associations rather than causes. Because the dataset is fictional, our results represent an academic demonstration and require validation on authorized operational data before business use.

6. Lecturer Feedback and Evaluation

We record lecturer feedback after the review and track each agreed change with an action, owner and completion status.

Feedback	Action	Owner	Status
Lecturer feedback	We record the agreed change and its completion evidence.	Assigned team member	Tracked
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7. Submission Evidence

Our final submission contains the planning, requirements and literature documents, Python analysis, SQL queries, model evaluation, Power BI dashboard, test evidence and final presentation. We confirm grading weights with the lecturer.